

Computer Science 2210 – Quiz 1

September 28, 2011

Grade: 89

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1. Regarding our class discussions of **collections** of **System.Object**, answer the following questions

- Name one advantage a **collection** of **System.Object** has all objects may share in this collection
- Name the major disadvantage of such a collection Must cast objects to correct type when necessary

2. Specify whether each of the following is **True** or **False**

- false A **generic** class is a template or blueprint from which many actual classes may be created
- true A **generic** class uses **placeholders** or **parameterized types** within the class definition to designate where concrete types will be substituted when the generic class is used
- true **List<T>** is an example of a generic collections class in the **.NET** library
- false If **Book** is a user-defined class, **List<Book>** is a **generic class**
- false If **L** is a **List<Employee>** where **Employee** is a class, **L.Capacity ≤ L.Count**
- true We discussed some **interfaces** that **List<T>** must implement so that its methods such as **Contains**, **Sort**, **IndexOf**, and **BinarySearch** will work properly
- true Existing items in a **List<Employee>** may be accessed by subscript, similar to an **array** of **Employee**
- false The first item in a **List** instance **L** is **L[1]**
- false The **Size** property of a **List** tells how many items are currently stored in the **List**
- true A **generic class** may have more than one **parameterized type**
- true **C#** allows some **operators** to be **overloaded** for user-defined classes
- true **Arithmetic operators** such as **+**, **-**, *****, and **/** may be defined for objects of a user-defined class
- true **Comparison operators** such as **<**, **>**, **==**, and **!=** may be defined for objects of a user-defined class
- false One may change the meaning of **operator** on a **primitive type** by **overloading the operator**
- true **Overloaded operators** must be **static**
- false A **static** component (method, attribute, property, ...) of a class is an **instance component** – i.e., every object of the class has one
- false One may create a **⊙ operator** in a user-defined class
- true If **p** is an object of the class **Person**, the class may contain an **overload** of the ***** **operator** so that the code **p = p * 8;** compiles and executes correctly

3. Is **Gismo** is a class that we are defining, and if the program using **Gismo** is to have and work with a fully functional **List<Gismo>**, the two interfaces that **Gismo** must implement are IComparable and IEquatable

- For the **interface** needed to support methods such as **Contains** and **IndexOf**, write the header line (first line) of each method that should be implemented. Be specific and give the full line for each. Example:

private double AddEmUp (double a, int b)

public bool Equals(T type)

public int GetHashCode()

public overrides bool Equals (Object o)

- For the interface needed to support methods such as **Sort**, write the header line (first line) of each method that should be implemented. Be specific and give the whole line for each.

public int CompareTo (T type)